Pandelis Zembashis

CTO | Agent Runtime Engineer | Self-Improving AI Systems

p@ndel.is | pa.ndel.is | github.com/PandelisZ | linkedin.com/in/Pandelis

EXECUTIVE SUMMARY

CTO and hands-on engineer building agent runtimes, evaluation loops, secure execution systems, and developer infrastructure for applied AI labs. At Cosine, built and led much of the platform behind AI coding agents: RL-style feedback loops on frontier closed models, 20,000+ coding gyms in a multi-month rollout, tool use, repository search, deterministic replay, subagents, sandboxed execution, and traces from real coding work. Strong fit for Recursive through practical experience building systems where agents improve through evaluation, feedback, search, parallel exploration, and safe execution rather than one-off prompting.

PROFESSIONAL EXPERIENCE

Cosine | CTO | Oct. 2023 – May 2026

Applied AI lab building coding agents, model feedback loops, secure execution, and enterprise deployment infrastructure.

- Led product and engineering for Cosine's AI coding platform, setting technical direction across agent architecture, tool use, context management, runtime reliability, developer experience, and deployment.
- Built the core Go agent harness for streaming tool calls, event scheduling, checkpoints, deterministic replay, memory writes, MCP/LSP integrations, subagents, file locks, repository search, browser use, and structured rollout traces.
- Built ML feedback loops around frontier closed-model agents, including RL-style training/evaluation signals from traces, tests, LSP diagnostics, reviews, conversation patterns, and parallel sandboxed runs.
- Led a multi-month rollout of 20,000+ coding gyms, turning real agent attempts into repeatable evaluation environments for improving tool use, planning, recovery, and long-horizon code modification.
- Designed workflow orchestration for multi-agent software engineering, including background subagents, task state propagation, human review, state recovery, and long-horizon execution.
- Worked across OpenAI, Anthropic, Gemini, and internal model capabilities, shaping prompts, reward/evaluation signals, tool contracts, context strategy, and provider-specific behavior into stable runtime patterns.
- Built secure remote execution for model-controlled coding work: Kubernetes workers, gVisor/runsc isolation, ephemeral sessions, WebSocket control planes, network boundaries, and private deployment paths.
- Helped position Cosine as a full-stack AI lab with benchmark-leading coding models, AI-agent infrastructure, and public recognition for sovereign AI deployment.

11x.ai | Head of Engineering | Apr. 2023 – Sep. 2023

Fastest-growing UK AI startup from 0 to $1M ARR in under six months. TechCrunch coverage.

- Scaled the codebase and engineering organization behind Alice as 11x.ai grew from 0 to $1M ARR in under six months.
- Scaled LLM-backed outbound automation systems that processed millions of email chains and billions of tokens, with event-driven architecture, worker orchestration, and operational monitoring.
- Established engineering cadence, onboarding, incident response habits, and cross-functional execution across product, engineering, and GTM during rapid growth.

Papaya | Technical Lead | Feb. 2022 – Mar. 2023

Marketplace and fleet platform helping electrify commercial and last-mile logistics. TechCrunch coverage.

- Led delivery of a production fleet management platform in Next.js, React Query, Tailwind CSS, and TypeScript.
- Modernized the Ruby on Rails backend by introducing GraphQL APIs, improving iteration across web, operations, and marketplace workflows.
- Established development cadences, product intake, and stakeholder feedback loops that improved execution across engineering and product.

Weezy (acquired by Getir) | Senior Software Engineer | Aug. 2021 – Jan. 2022

Rapid grocery delivery startup building warehouse, ordering, and customer systems.

- Took over the core ordering and processing monolith in Ruby on Rails, owning day-to-day delivery and forward planning during a high-growth phase.
- Built warehouse operations systems and supporting AWS and Heroku infrastructure with a focus on scale, availability, incident response, and security.

finleap connect | Senior Software Engineer | Nov. 2020 – Aug. 2021
White-label fintech platform powering digital banking products and apps.

- Built GraphQL APIs on NestJS and cross-platform clients in React Native, Apollo, Node.js, and TypeScript for a multi-tenant fintech platform.
- Led alerting and observability improvements while supporting migration from ECS to Amazon Lambda.
- Owned client-specific delivery across the stack, from GraphQL resolvers and mutations to customer-facing mobile screens.

IoT CONNCTD GmbH | Full Stack Engineer | Jul. 2019 – Nov. 2020
Cloud IoT platform unifying device protocols behind a GraphQL API.

- Built frontend interfaces, SDKs, and internal tooling for a cloud IoT platform exposing a unified GraphQL API across multiple device protocols.
- Designed authentication, update, and upstream synchronization flows using TypeScript, React, MobX, and Apollo GraphQL.

SELECTED OSS, WRITING & PUBLIC SIGNALS

Cosine Agent Systems: Why we built our own harness; Cosine Swarm; Parallelising software development.
Sovereign AI Coverage: Tech.eu on Cosine: benchmark-leading coding models, agent infrastructure, air-gapped deployment, and UK sovereign AI strategy.
OSS: claude-agent-sdk-rust; claude-agent-sdk-go; codex-web-sdk; gut; jalla; 100+ merged public GitHub PRs.
Public Material: pa.ndel.is; Devpost portfolio; AWS Startups - The Future Of Work.

EDUCATION

Birmingham City University | Bachelor's Degree, Computer Science with Year in Industry | 2015 – 2019

AREAS OF EXPERTISE

Agent Systems: Self-improving coding agents, tool use, long-horizon tasks, subagents, context management, memory, MCP/LSP
Evaluation & Feedback Loops: RL-style coding gyms, deterministic replay, rollout traces, test/LSP signals, parallel exploration, failure analysis
Runtime & Secure Execution: Go, Rust, TypeScript, Kubernetes, gVisor/runsc, WebAssembly, WebSockets, sandboxing, repository-scale search
Applied ML Infrastructure: OpenAI, Anthropic, Gemini, provider abstractions, prompt/tool contracts, research-to-product loops, observability
Leadership: CTO-level ownership, small-team execution, technical roadmaps, mentoring, product judgment, enterprise deployment